

CBCS SCHEME

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BCV654C

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Integrated Waste Management for a Smart City

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Briefly explain the physical and chemical characteristics of solid waste.	10	L2	CO1
	b.	Explain in detail about collection of municipal solid waste.	10	L2	CO1
OR					
Q.2	a.	Explain briefly about the future challenges and issues with respect to integrated solid waste management.	10	L2	CO1
	b.	Define sanitary landfill. Explain the factors to be considered for the selection of site.	10	L2	CO1
Module – 2					
Q.3	a.	Define Composting. Explain the factors affecting the composting.	10	L2	CO2
	b.	Explain aerobic and anaerobic composting with its micro biology.	10	L2	CO2
OR					
Q.4	a.	Explain briefly the salient features of Solid Waste Management Rules 2016.	10	L2	CO2
	b.	Explain briefly the vision, mission and objectives of Swachh Bharath Mission.	10	L2	CO2
Module – 3					
Q.5	a.	List the methods of energy recovery system and explain any one with neat flow chart.	10	L2	CO3
	b.	Briefly explain the current issues of solid waste management in India.	10	L2	CO3
OR					
Q.6	a.	List any 10 smart cities proposed in first list of smart city programme and briefly explain the status of waste management of any two smart cities.	10	L2	CO3
	b.	Briefly explain incineration and pyrolysis with a neat sketch.	10	L2	CO3
Module – 4					
Q.7	a.	Briefly explain C and D waste. Enlist the activity that generate C and D waste and its effects.	10	L2	CO4
	b.	Explain briefly C and D Waste Rules 2016.	10	L2	CO4

OR

Q.8	a.	Briefly explain the beneficial use of C and D waste.	10	L2	CO4
	b.	List the hazardous material in C and D waste and explain any two hazardous material in C and D waste.	10	L2	CO4

Module – 5

Q.9	a.	Briefly explain the current issues and challenges faced by e-waste management in India.	10	L2	CO4
	b.	Briefly explain the E-waste Management Rules 2016.	10	L2	CO4

OR

Q.10	a.	Briefly explain the sources of e-waste and its effects on human health.	10	L2	CO4
	b.	Explain any two methods for disposal of e-waste.	10	L2	CO4

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